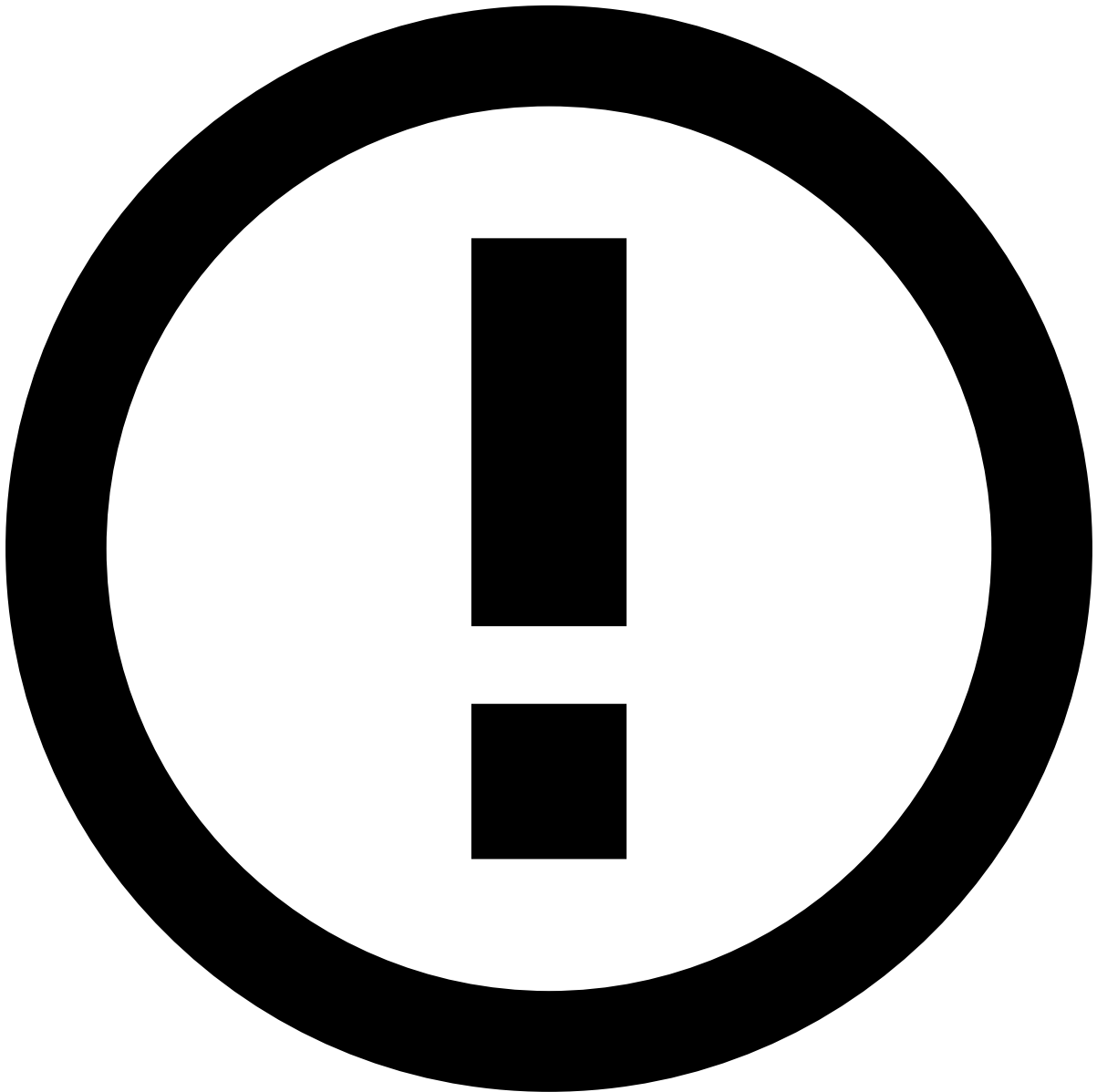


On this page

Dead Letters System



Dead letter message loggers

To refer to the old dead letter system, see [Dead letter messages logger](#).

Introduction

It's possible to configure different dead letters systems for the different features of Case Manager. These dead letters are used to make sure CM doesn't lose important information, such as getting an event from the queue and encountering an error that prevents the event from being properly processed and persisted in Case Manager.

This dead letter system supports encryption, so that confidential data remains protected. It also has auto-recovery, in order to recover dead letters without any human intervention.

There are several dead letters in Case Manager. The following list displays the available dead letters that can be configured according to the needs of the system. Note that some share the same configuration since they are very similar (e.g.: batches dead letters):

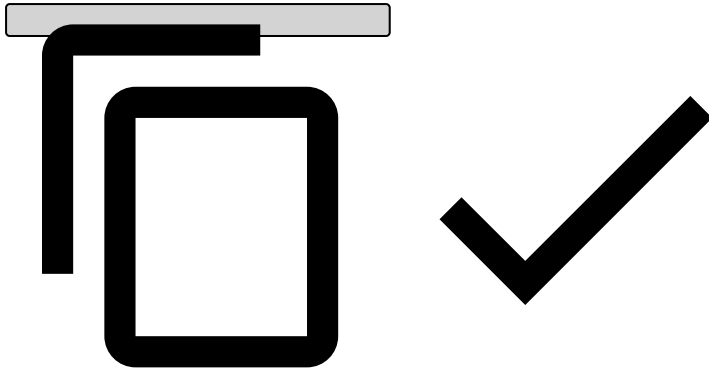
- **EVENT_QUEUES**, for errors during event ingestion.
- **PULSE_FEEDBACKS**, for errors while sending status change outcomes (feedback) to Pulse.
- **AUTOMATION_MANAGER**, for automation rules errors (e.g. failure to execute an automation due to an event not being persisted yet by the event storage batch).
- **EVENT_STORAGE**, for errors while persisting events to the AM_EVENT_STORAGE table.
- **EXPLANATIONS_STORAGE**, for errors while persisting events to the AM_EVENT_EXPLANATION table.
- **AUTOMATION_STORAGES**, for errors while persisting the results of automation rule actions to the different tables.
- **EVENT_COMMON_STORAGE**, for errors while persisting events to the AM_EVENT_COMMON table.

Core Configuration

To enable this feature, add it to the configuration as displayed in the following example:

config.json

```
...
"deadletter": {
  "confs" {
    "event_queues": {
      "enabled": true,
      "dump_directory": "/some-directory",
      "flushing_interval": 5000,
      "max_recovery_attempts": 3,
      "queue_size": 1000,
      "recovery_cron_expression": "0 0 * ? * * *",
      "rate_limiter": 0,
      "fatal_failure_regex": ".*IllegalStateException.*",
      "encryption_dump_config": {
        "enabled": false,
        "secret": "815a829f2a918959822dc33080a8a58c",
        "cipher_mode": "AES/GCM/NoPadding"
      }
    }
    "recovery_job_timeout": 7200000
  },
  "feedback": {
    // Same as "event_queues".
  },
  "triggers": {
    // Same as "event_queues".
  },
  "storage_batches": {
    // Same as "event_queues".
  }
}
},
...
```



There are 4 main groups of configurations:

- **event_queues**: Applies to the **EVENT_QUEUES** dead letter.
- **feedback**: Applies to the **PULSE_FEEDBACKS** dead letter.
- **triggers**: Applies to the **AUTOMATION_MANAGER** dead letter.
- **storage_batches**: Applies to all the batch storage related dead letters, namely, **EVENT_STORAGE**, **EXPLANATIONS_STORAGE**, **AUTOMATION_STORAGES** and **EVENT_COMMON_STORAGE**.

The configuration for each of the groups is exactly the same, but must be done separately.

Dead Letter Configuration

Configuration Key	Description	Default	Restrictions
enabled	Whether the dead letter system is enabled.	false	true, false
dump_directory	The directory where the dead letter folders will be created.	var/dead-letters	String
flushing_interval	The flush interval <i>in milliseconds</i> used to flush the writers to disk.	1000 (1 second)	> 0
max_recovery_attempts	The maximum number of times that a dead letter can be reprocessed.	3	>= 0 (0 means no recovery)
queue_size	The maximum size for the in memory queues where dead letters are persisted before being written to disk.	1000	> 0
recovery_cron_expression	The recovery cron expression that defines when the auto-recovery will run.	0 0 * ? * * (every 1 hour)	Any valid cron expression
rate_limiter	The amount of entries that can be recovered per second.	0 (unlimited)	>= 0
fatal_failure_regex	The regex to detect fatal failures which will mark an entry as non-recoverable.	<i>empty string</i>	Any valid regex
encryption_dump_config	The configurations necessary to provide encryption support to the dead letter files.	Check encryption table	Disabled by default
recovery_job_timeout	The timeout (maximum amount of time) that the recovery job can take <i>in milliseconds</i> before being aborted.	7200000 (2 hours)	> 0

Dead Letter Encryption Configuration

Configuration Key	Description	Default	Restrictions
enabled	Whether the encryption is enabled.	false	true, false
secret	The secret key encrypted and encoded in hexadecimal.	<i>empty string</i>	String
cipher_mode	The cipher mode to be used.	"AES/GCM/NoPadding"	Valid Cipher Mode

Cluster Configuration

Configuration Key	Description	Default	Restrictions
enabled	Whether the cluster-mode is enabled.	false	true, false
database_driver	The JDBC Driver class name.	"org.postgresql.Driver"	String
validation_query	Query used to validate the connection each 50s.	empty string	String

Auto-Recovery

The auto-recovery executes based on the *recovery_cron_expression*. For each of the contexts, it performs a different recovery action, namely:

- **EVENT_QUEUES**: events are re-injected into the RabbitMQ queue.
- **PULSE_FEEDBACKS**: feedbacks are re-sent to Pulse.
- **AUTOMATION_MANAGER**: triggers are re-processed by the Automation Manager.
- **EVENT_STORAGE**, **EXPLANATIONS_STORAGE** and **EVENT_COMMON_STORAGE**: models are re-added to the batch.
- **AUTOMATION_STORAGES**: if it fails to be persisted, it will be re-added to the batch. If it fails after being persisted, it will re-run the post flush actions.

Monitoring Metrics

The dead letter system contains multiple metrics to monitor what is happening. The metrics are the same for all the contexts but are prefixed differently depending on the context.

All the metrics start with `cm.<context-metric-prefix>`, whether *context-metric-prefix* changes depending on the context:

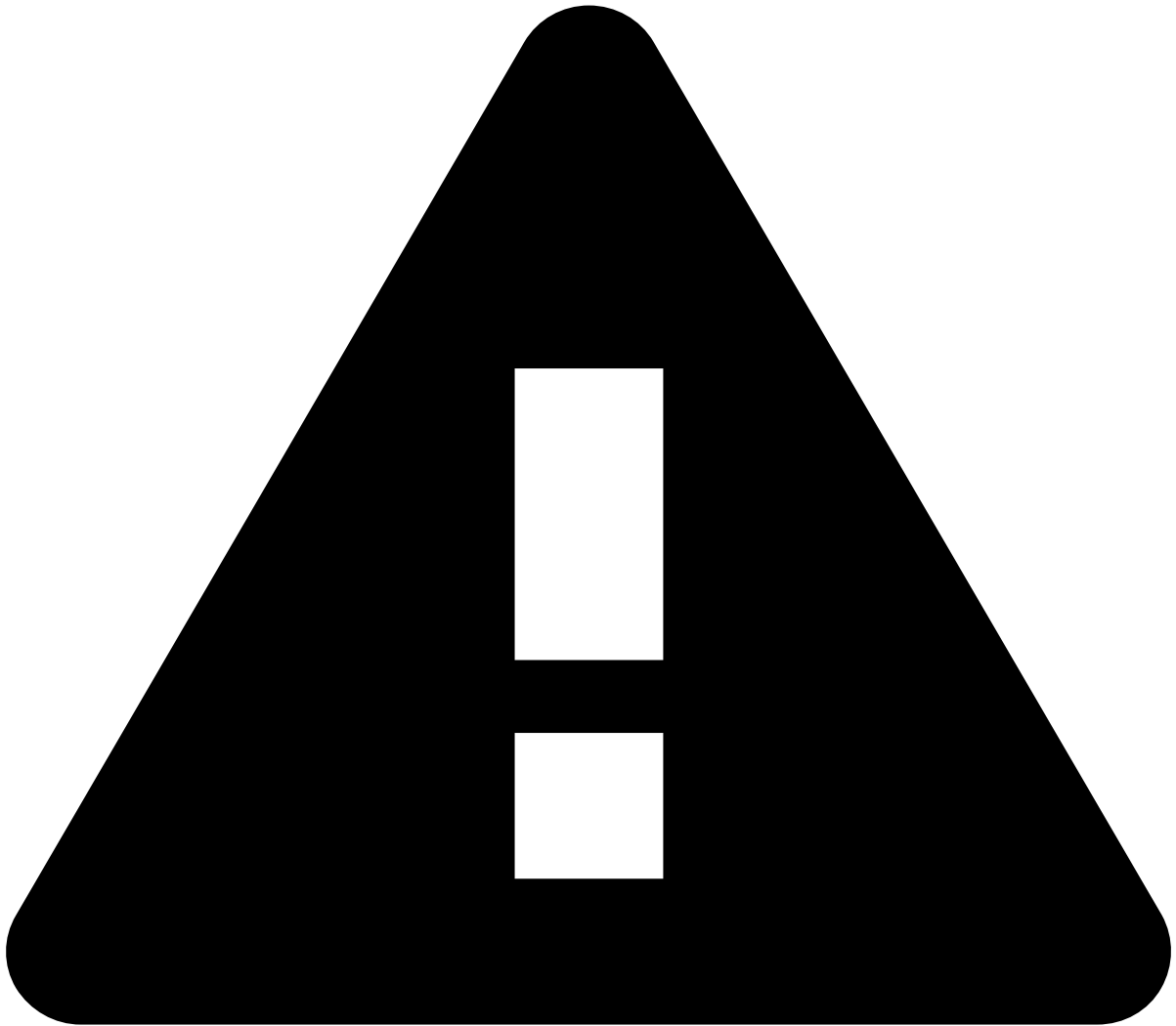
- **EVENT_QUEUES**: `event_queues`
- **PULSE_FEEDBACKS**: `feedback`
- **AUTOMATION_MANAGER**: `automation_manager`
- **EVENT_STORAGE**: `event_storage`
- **EXPLANATIONS_STORAGE**: `explanations_storage`
- **AUTOMATION_STORAGES**: `automation_storage`
- **EVENT_COMMON_STORAGE**: `event_common_storage`

<code>cm.<context-metric-prefix>.dump.storage.retry.queue.size</code>	The total number of dead letters that are present in the in-memory queue to be written to the dump file in the retry folder.	<code>appType=CASE_MANAGER</code> <code>metricType=APPLICATION</code> <code>batch={BATCH_NAME}</code> (only applies to <i>AUTOMATION_STORAGES</i> metrics) <code>channel={QUEUE_NAME}</code> (only applies to <i>EVENT_QUEUES</i> metrics)	Gauge
<code>cm.<context-metric-prefix>.dump.storage.failed.queue.size</code>	The total number of dead letters that are present in the in-memory queue to be written to the dump file in the failed folder.	<code>appType=CASE_MANAGER</code> <code>metricType=APPLICATION</code> <code>batch={BATCH_NAME}</code> (only applies to <i>AUTOMATION_STORAGES</i> metrics) <code>channel={QUEUE_NAME}</code> (only applies to <i>EVENT_QUEUES</i> metrics)	Gauge

cm.<context-metric-prefix>.dump.storage.recovery.job	Whether the auto recovery job, which reprocesses dead letters dumped into the retry folder, is running or not.	appType=CASE_MANAGER metricType=APPLICATION batch={BATCH_NAME} <i>(only applies to AUTOMATION_STORAGES metrics)</i> channel={QUEUE_NAME} <i>(only applies to EVENT_QUEUES metrics)</i>	Gauge
cm.<context-metric-prefix>.dump.storage.dump.entries.retry.counter	The total number of dead letters that are currently stored in the disk to be retried by the auto recovery job.	appType=CASE_MANAGER metricType=APPLICATION batch={BATCH_NAME} <i>(only applies to AUTOMATION_STORAGES metrics)</i> channel={QUEUE_NAME} <i>(only applies to EVENT_QUEUES metrics)</i>	Counter
cm.<context-metric-prefix>.dump.storage.dump.entries.failed.counter	The total number of dead letters that are currently stored in the disk but already reached the maximum amounts of retry. These dead letters are no longer reprocessed and need to be recovered manually.	appType=CASE_MANAGER metricType=APPLICATION batch={BATCH_NAME} <i>(only applies to AUTOMATION_STORAGES metrics)</i> channel={QUEUE_NAME} <i>(only applies to EVENT_QUEUES metrics)</i>	Counter

Clustering

If enabled, the dead letter system creates a cluster with all Case Manager instances, which means that the instances are able to recover dead letters from other instances. This mode is important for deployments that need external persisted volumes (e.g.: kubernetes).



caution

To make sure that all dead letters can be recovered, it is important to have a shared filesystem where the dead letter can be written.

To enable cluster-mode, add the following configuration to the `config.json` file:

```
"deadletter": {  
  "confs" {  
    ...  
    ...  
  },  
  "clustering": {  
    "enabled": "true",  
    "database_driver": "org.database.TheDriver";  
    "validation_query": "select * from TABLE limit 1;  
  }  
},
```

